# Synchronization

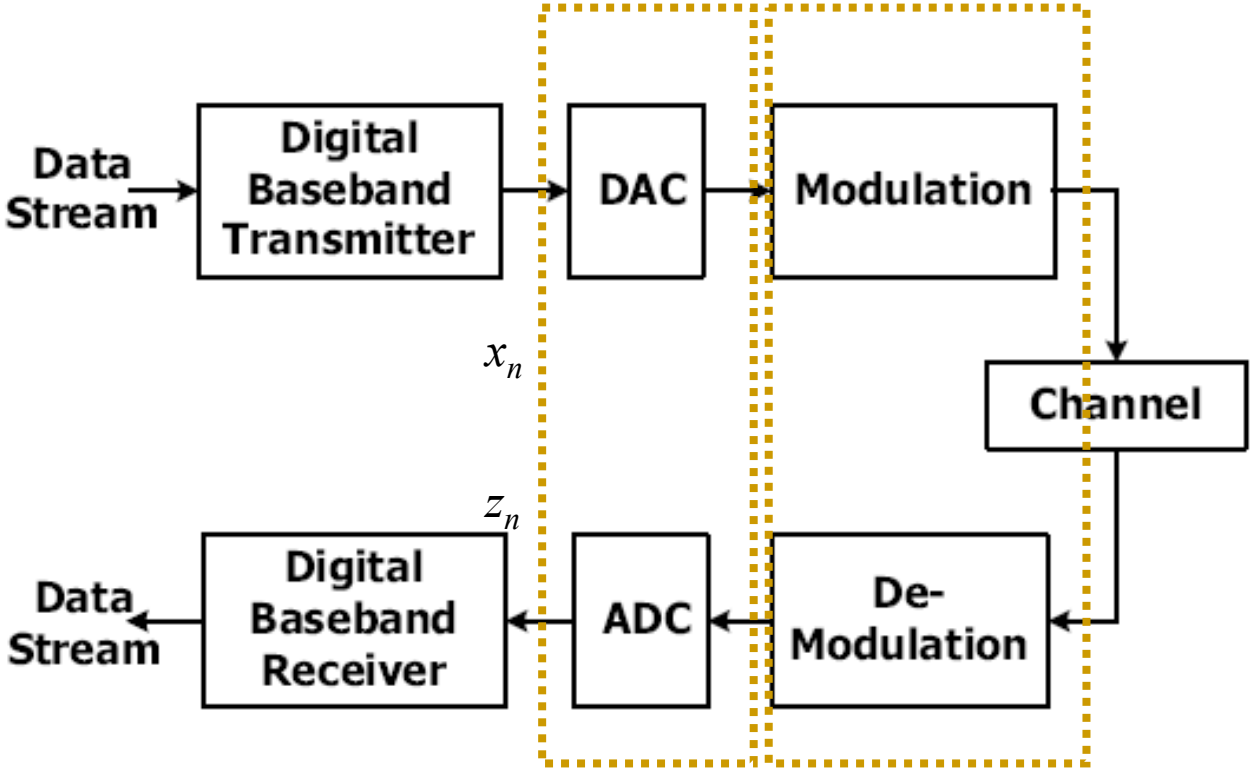
Instructor : Pei-Yun Tsai

Digital Communication IC Design

"Baseband Receiver Design for Wireless MIMO-OFDM Communications" 2nd Edition, Wiley 2012.
Chapter 6 Synchronization

# Source of Synchronization Error

- Oscillator mismatch
- Doppler effect
- Phase noise



# Synchronization Recovery

- Carrier recovery
  - To recover the carrier signal, $\cos(2\pi f_c t)$ , from the received modulated signal.
- Timing recovery
  - To find the correct timing for sample-and-hold or integrate-and-dump devices in the receiver.

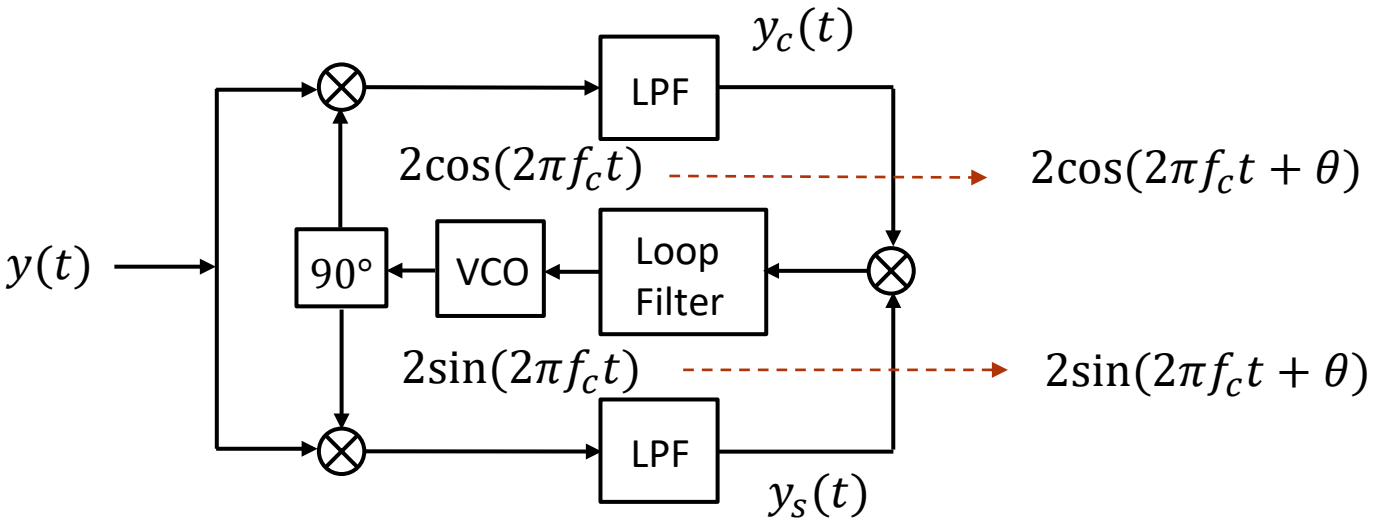
# I. Synchronization for Single Carrier System

# Carrier Recovery

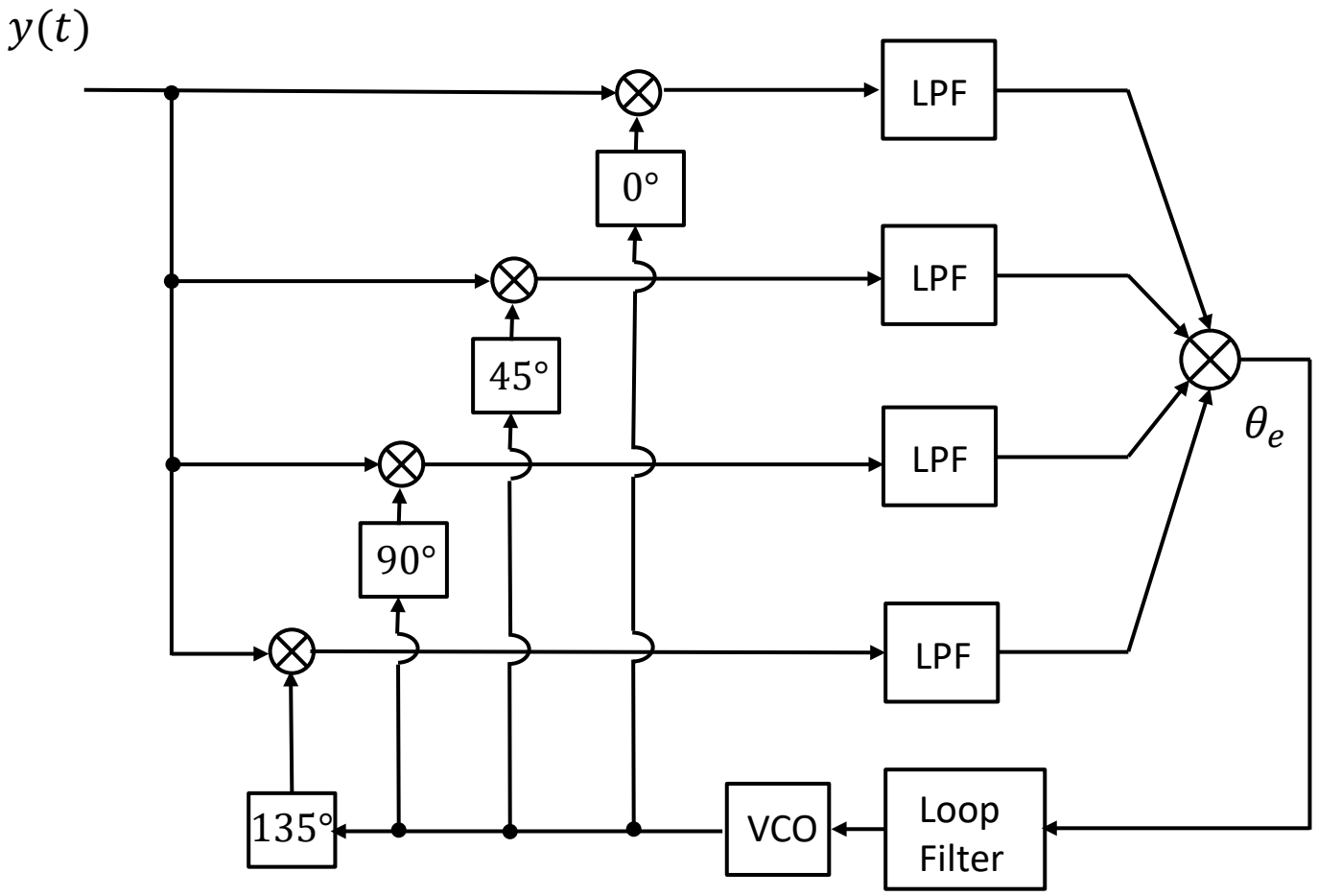
- To recover the carrier signal,$\cos(2\pi f_c t)$, from the received modulated signal $A_k \cos(2\pi f_c t + \phi_k)$.

# BPSK Costas Loop

- Assume that the phase error is $\epsilon$, and the base-band received signal is
  - $y(t) = \cos(2\pi f_c t + \phi_k + \epsilon) + n(t),$
- Ignore the noise, then
  - $y_c(t) = \cos(\phi_k + \epsilon), y_s(t) = \sin(\phi_k + \epsilon),$
  - $z(t) = y_c(t)y_s(t) = \cos(\phi_k + \epsilon)\sin(\phi_k + \epsilon) = 0.5 \cdot \sin(2\phi_k + 2\epsilon)$



# QPSK Costas Loop



# II. Synchronization Error for OFDM Systems

# OFDM System Model

- Frequency-domain data at the *k*th subcarrier of the *i*th symbol: $X_{i,k}$
- Time-domain sample n of the *i*th symbol:$x_{i,n}$

$$x(i(N+N_g)T_s + N_g\ T_s + nT_s) = x_{i,n} = \frac{1}{N}\sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} X_{i,k} e^{j2\pi nk/N}$$

- Multipath fading channel

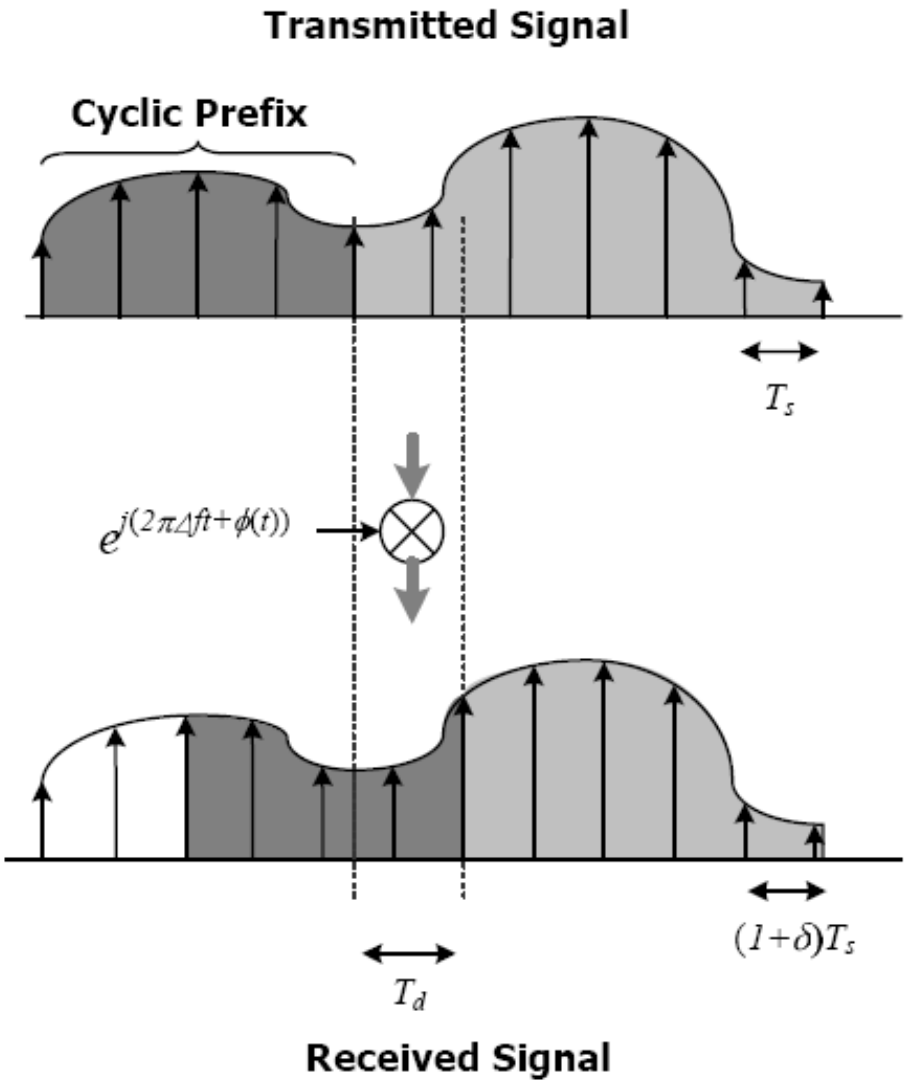
$$h(t) = \sum_r h_r \cdot \delta(t - \tau_r)$$

- Received time-domain signal

$$z(t) = \sum_r h_r \cdot x(t - \tau_r) + v(t)$$

# Synchronization Error

- Carrier frequency offset
- Carrier phase error
- Sampling clock offset
- Symbol timing offset

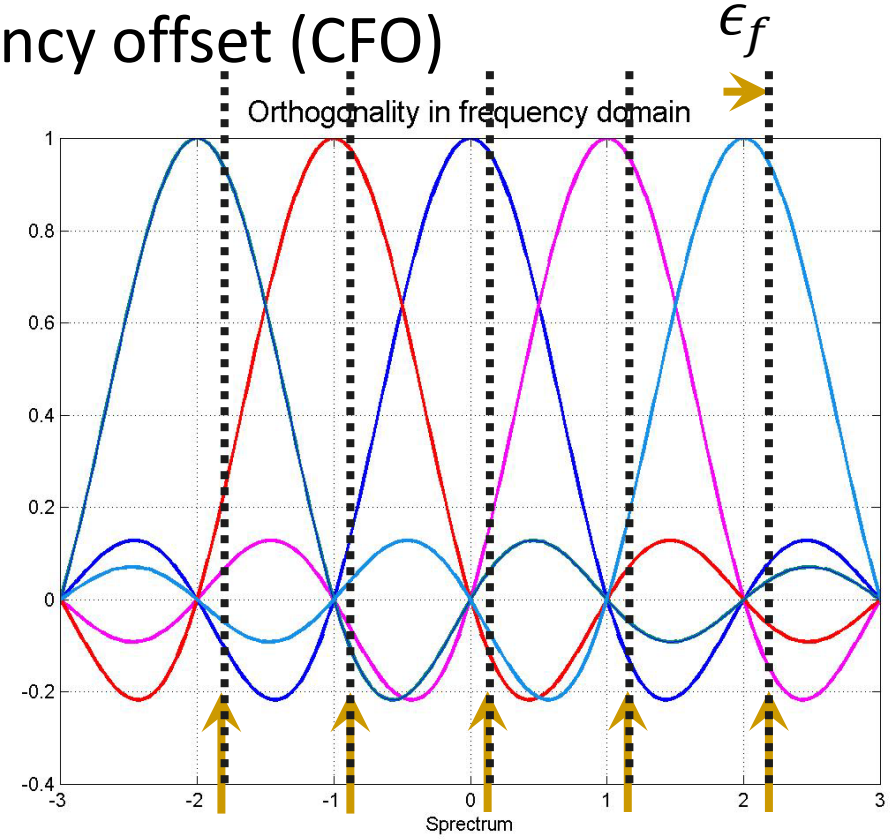


# Carrier Frequency Offset (1/2)

- Received baseband signal becomes

$$z_{i,n} = z(t)e^{j2\pi\Delta f t}\Big|_{t=i(N+N_g)T_s+\ N_gT_s+nT_s}$$

- Normalized carrier frequency offset (CFO)
  - Fractional CFO $\epsilon_f$
  - Integer CFO $\epsilon_I$



# Carrier Frequency Offset (2/2)

- Frequency-domain signal

$$Z_{i,k} = X_{i,k-\epsilon_I} H_{k-\epsilon_I} \frac{\sin(\pi\epsilon_f)}{N\sin(\frac{\pi\epsilon_f}{N})} e^{j2\pi\frac{i(N+N_g)+N_g}{N}(\epsilon_I+\epsilon_f)} e^{j\pi\frac{N-1}{N}(\epsilon_f)}$$
$$+ \sum_{l=-\frac{N}{2}, l\neq k-\epsilon_I}^{\frac{N}{2}-1} X_{i,l} H_l \frac{\sin(\pi(\epsilon_I+\epsilon_f+l-k))}{N\sin(\frac{\pi(\epsilon_I+\epsilon_f+l-k)}{N})} e^{j2\pi\frac{i(N+N_g)+N_g}{N}(\epsilon_I+\epsilon_f)} e^{j\pi\frac{N-1}{N}(\epsilon_I+\epsilon_f+l-k)}$$

- Influences
  - Index shift
  - Magnitude attenuation
  - Identical phase shift
  - ICI

# Carrier Phase Offset

- Phase noise and phase offset $e^{j(\phi_{i,n}+\phi_0)}$
- Frequency-domain signal

$$Z_{i,k} = X_{i,k}H_k e^{j\phi_0}\left(\frac{1}{N}\sum_{n=0}^{N-1} e^{j\phi_n}\right) + \sum_{l=-\frac{N}{2},l\neq k}^{\frac{N}{2}-1} X_{i,l}H_l\, e^{j\phi_0}\left(\frac{1}{N}\sum_{n=0}^{N-1} e^{j\phi_{i,n}} e^{j2\pi n(k-l)/N}\right)$$

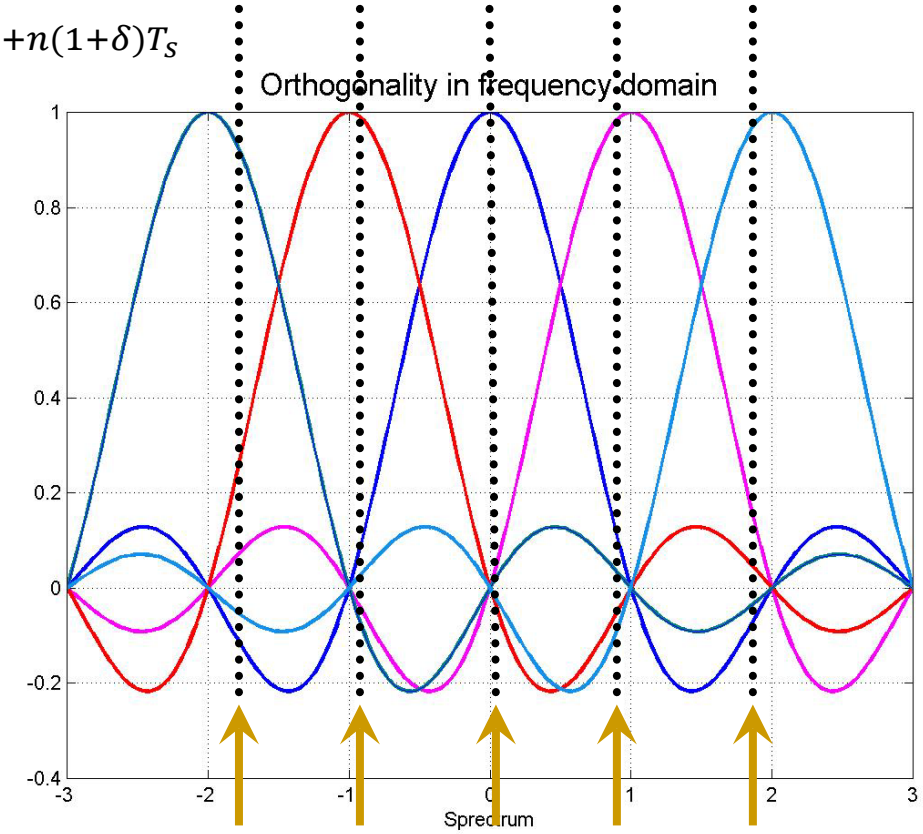
- Influences
  - Phase shift
  - ICI

# Sampling Clock Offset (1/2)

- Received signal becomes

$$z_{i,n} = z(t)\Big|_{t=i(N+N_g)(1+\delta)T_s+\ N_g(1+\delta)T_s+n(1+\delta)T_s}$$

- Large deviation to high frequency sub-carriers



# Sampling Clock Offset (2/2)

- Frequency-domain signal

$$Z_{i,k} = X_{i,k}H_k \frac{\sin(\pi\delta k)}{N\sin(\frac{\pi\delta k}{N})} e^{j2\pi\frac{i(N+N_g)+N_g}{N}\delta k} e^{j\pi\frac{N-1}{N}(\delta k)} + \sum_{l=-\frac{N}{2}, l\neq k-\epsilon_I}^{\frac{N}{2}-1} X_{i,l}H_l \frac{\sin(\pi((1+\delta)l-k))}{N\sin(\frac{\pi((1+\delta)l-k)}{N})} e^{j2\pi\frac{i(N+N_g)+N_g}{N}(\delta l)} e^{j\pi\frac{N-1}{N}((1+\delta)l-k)}$$

- Influences
  - Magnitude attenuation
  - Phase shift proportional to *k*
  - ICI

# Symbol Timing Offset

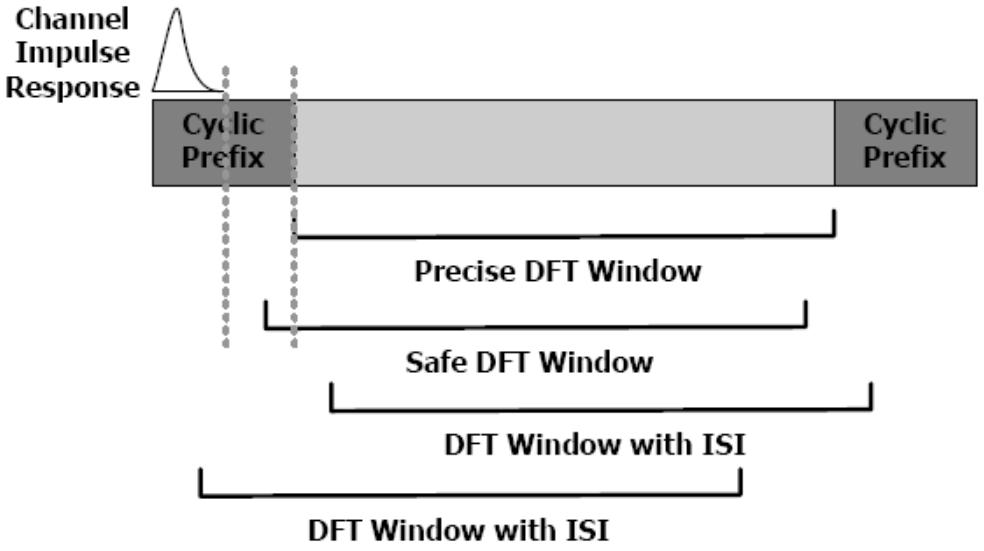
- Frequency-domain data without ISI

$$Z_k = X_k H_k e^{-j2\pi T_d k/(NT_s)} + V_k$$

- Frequency-domain data with ISI

$$Z_k = X_k H_k \frac{N - |T_d|/T_s}{N} e^{-j2\pi T_d k/(NT_s)} + ISI + V_k$$

- Influences if exists
  - ISI
  - Magnitude attenuation



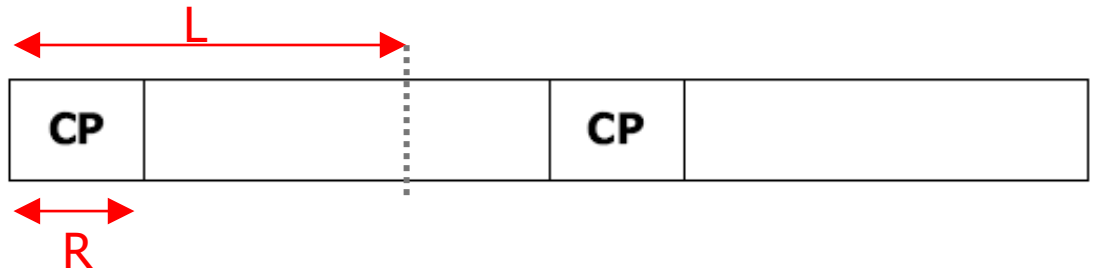
# Main Concerns

- Where to estimate & Where to compensate?
  - Time domain
  - Frequency domain
- Concerns
  - Transmission type (packet-based v.s. frame-based)
  - System resources
  - Latency
  - Performance

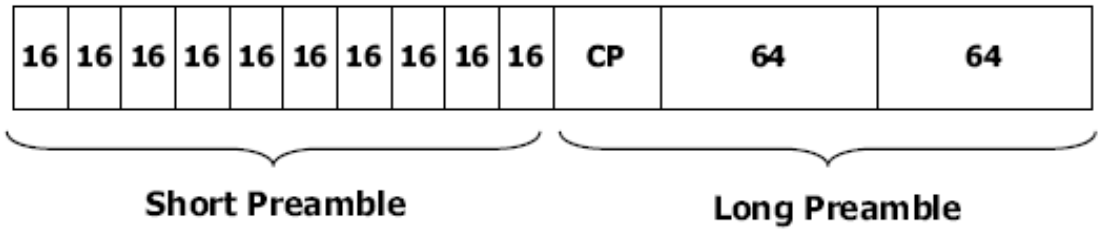
# Synchronization Algorithm for OFDM Systems

# Cyclic Repetitions

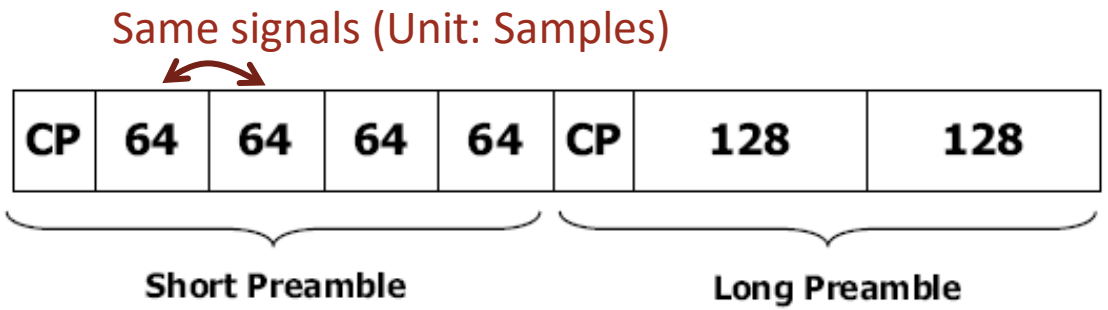
General OFDM Symbol




Preamble of 802.11a




Preamble of 802.16e



# Coarse Symbol Timing Detection

- Delay and correlate

$$\Phi_{DC}(m) = \left| \sum_{r=0}^{R-1} z_{m-r} \cdot z_{m-r-L}^{*} \right|$$

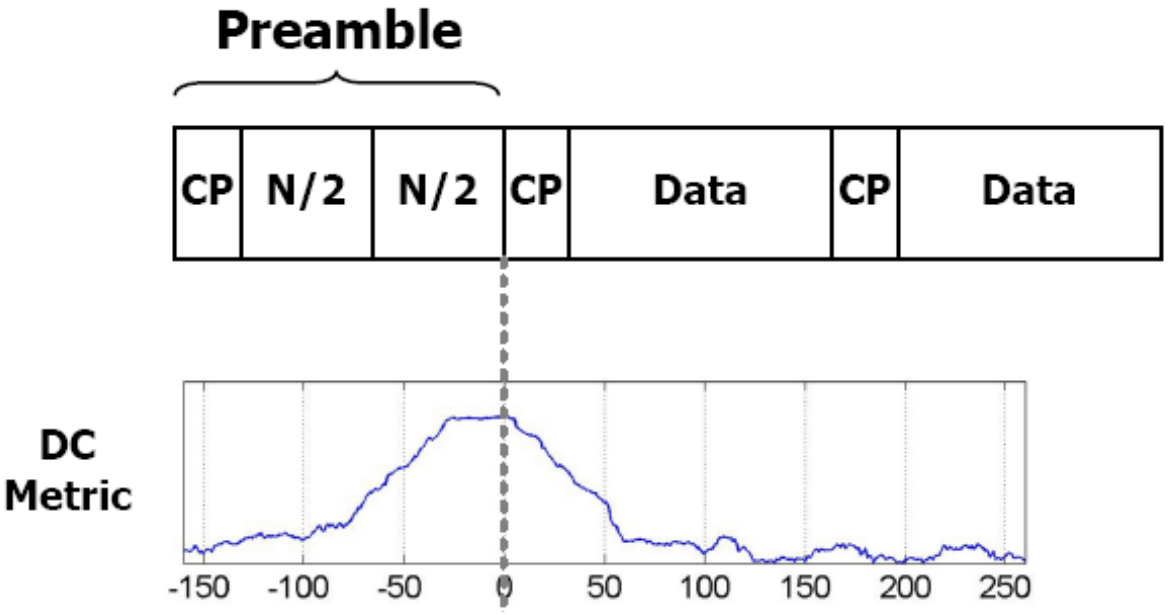
$$\hat{m}_{DC} = \arg max_m \, \Phi_{DC}(m)$$

- Normalized metric

$$\Phi_s(m) = \frac{\left| \sum_{r=0}^{R-1} z_{m-r} \cdot z_{m-r-L}^{*} \right|^2}{\left( \sum_{r=0}^{R-1} |z_{m-r}|^2 \right)^2}$$

# Output of Symbol Timing Detection

- L=N/2, R=N/2



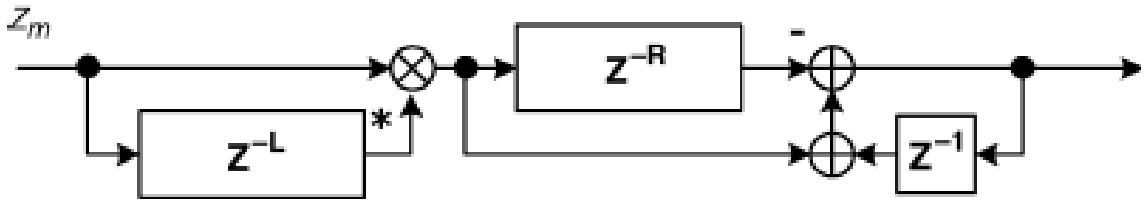
# Recursive Form for Implementation

- Recursive form

$$\sum_{r=0}^{R-1} z_{m+1-r} \cdot z_{m+1-r-L}^{*} = \left( \sum_{r=0}^{R-1} z_{m-r} \cdot z_{m-r-L}^{*} \right)$$

$$+ z_{m+1-r} \cdot z_{m+1-r-L}^{*} - z_{m-R+1} \cdot z_{m-R+1-L}^{*}$$

- Implementation



# Fine Symbol Timing Detection (1/2)

- Cross-correlation

$$\Phi_{zp}(m) = \left| \sum_{q=0}^{Q-1} z_{m+q} \cdot p_q^* \right|$$

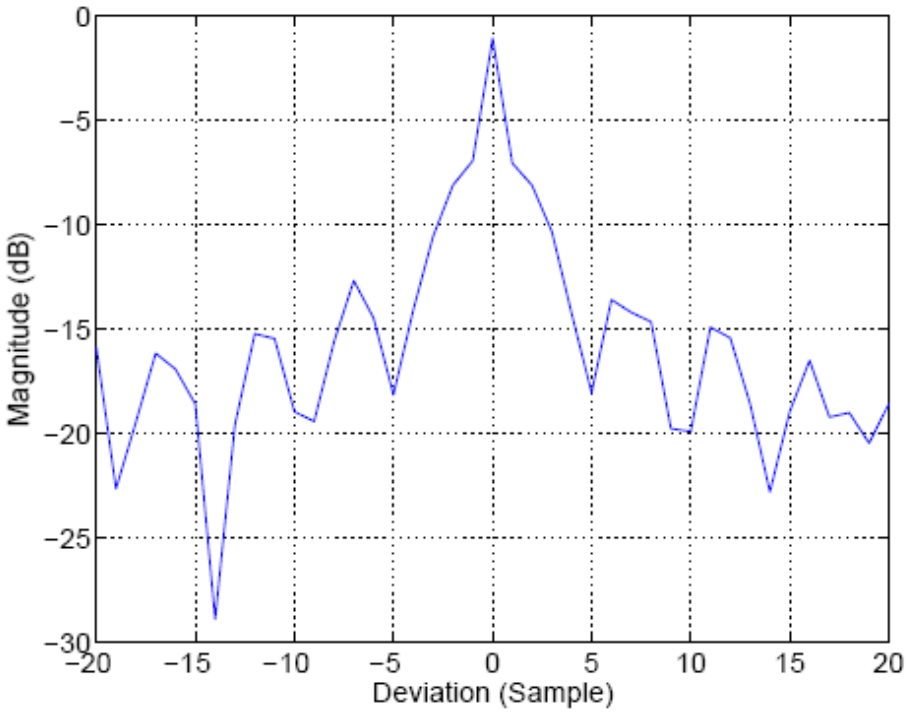
$$\hat{m}_{MAX} = \arg max_m \, \Phi_{zp}(m)$$

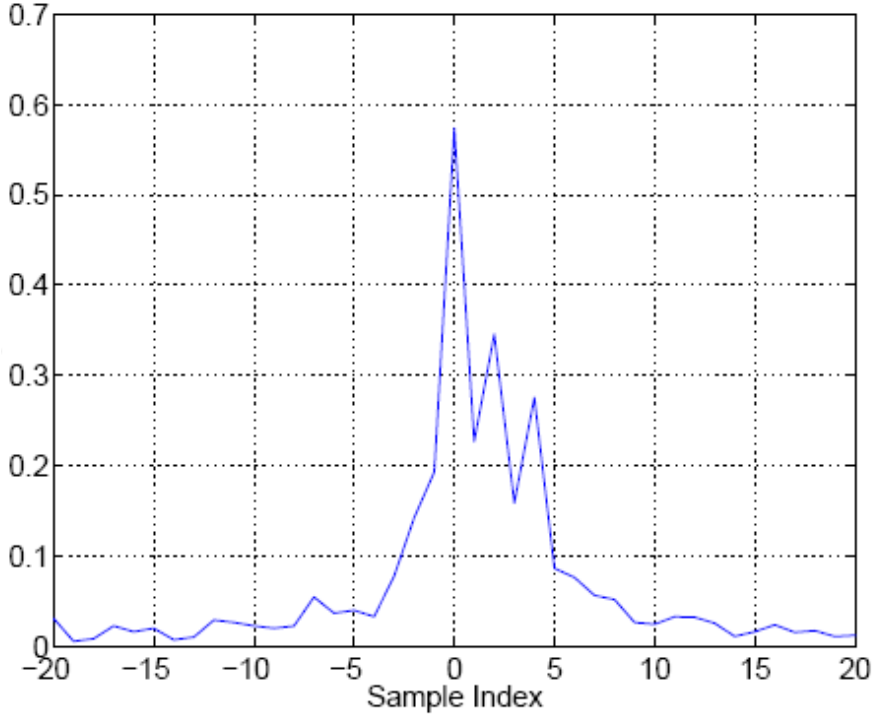
- Frequency-domain phase shift

$$\hat{m}_{PS} = -\frac{N}{2\pi} \measuredangle \left( \sum_k (Z_{k+1} Z_k^*) \, (X_{k+1} X_k^*)^* \right)$$

# Fine Symbol Timing Detection (2/2)

- Autocorrelation property v.s. matched filter output



# Fractional CFO Estimation

- Fractional CFO estimation can be accomplished when the symbol boundary is detected.
- The acquisition range of $\epsilon_f$ depends on the distance of repetitions.

$$\widehat{\Delta f} = \frac{1}{2\pi L T_s} \angle \left( \sum_{r=0}^{R-1} z_{m-r} z_{m-r-L}^{*} \right)$$

$$\widehat{\Delta f} = \widehat{\epsilon_f} f_{sub}$$



# Frequency Ambiguity

- Range for estimation

$$[-1/(2LT_s), 1/(2LT_s)]$$

- Example
  - DVB-T (L=N)

$$-0.5f_{sub} \leq \widehat{\Delta f} \leq 0.5f_{sub}$$

  - 802.16e-2005 (L=N/4)

$$-2f_{sub} \leq \widehat{\Delta f} \leq 2f_{sub}$$

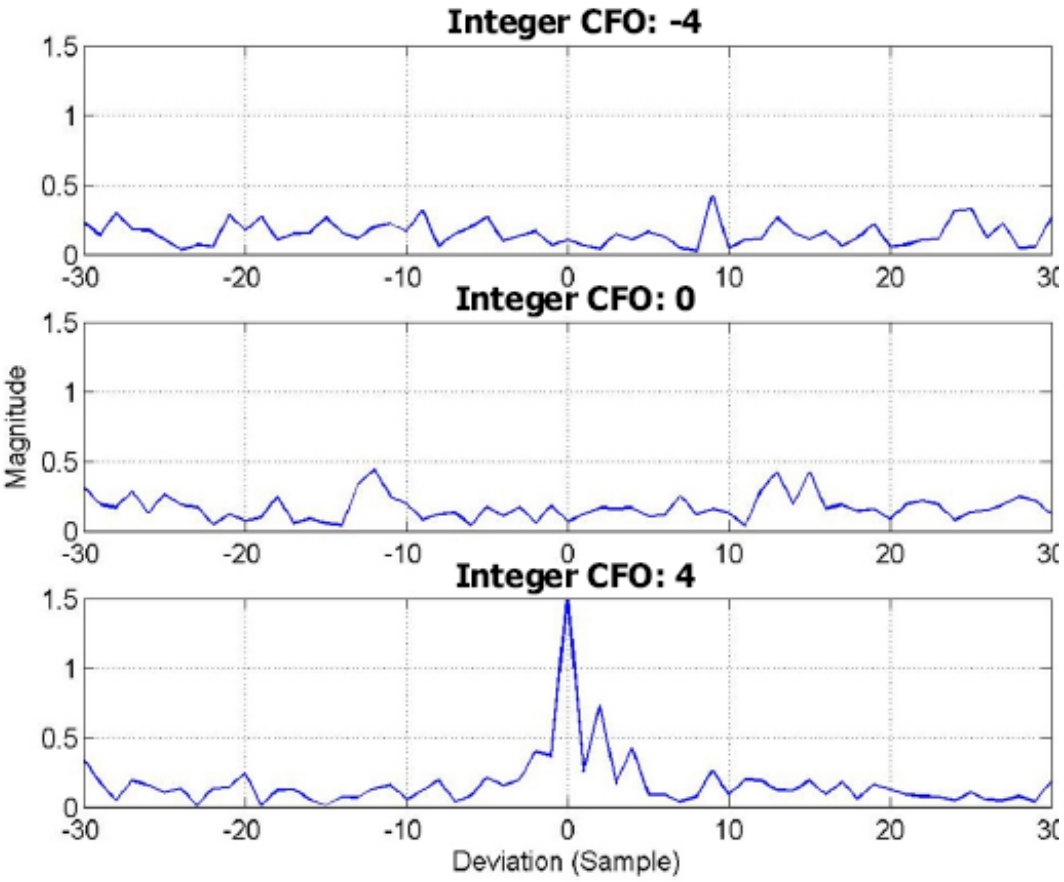
$f_{sub} = \frac{1}{NT_s}$ : subcarrier spacing

# Integer CFO

- Maximum CFO in 802.16e
  - Oscillator deviation: ±8ppm
  - Highest carrier frequency: 10.68 GHz
  - CFO: ±171KHz ~= 11 subcarrier spacing
  - With $-2f_{sub} \leq \widehat{\Delta f} \leq 2f_{sub}$, estimation for ±12, ±8 and ±4 subcarrier spacing is required

# Integer CFO Estimation (1/2)

- Time-domain correlation
  - CFO of 4.2 $f_s$
  - Matched filter with coefficients multiplied by various integer CFO

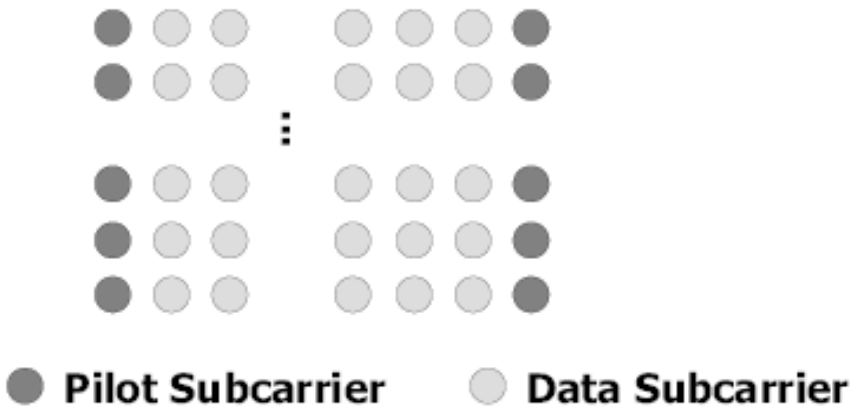


# Integer CFO Estimation (2/2)

- Frequency-domain autocorrelation
  - Take advantage of pilot subcarriers with index $\alpha_p$

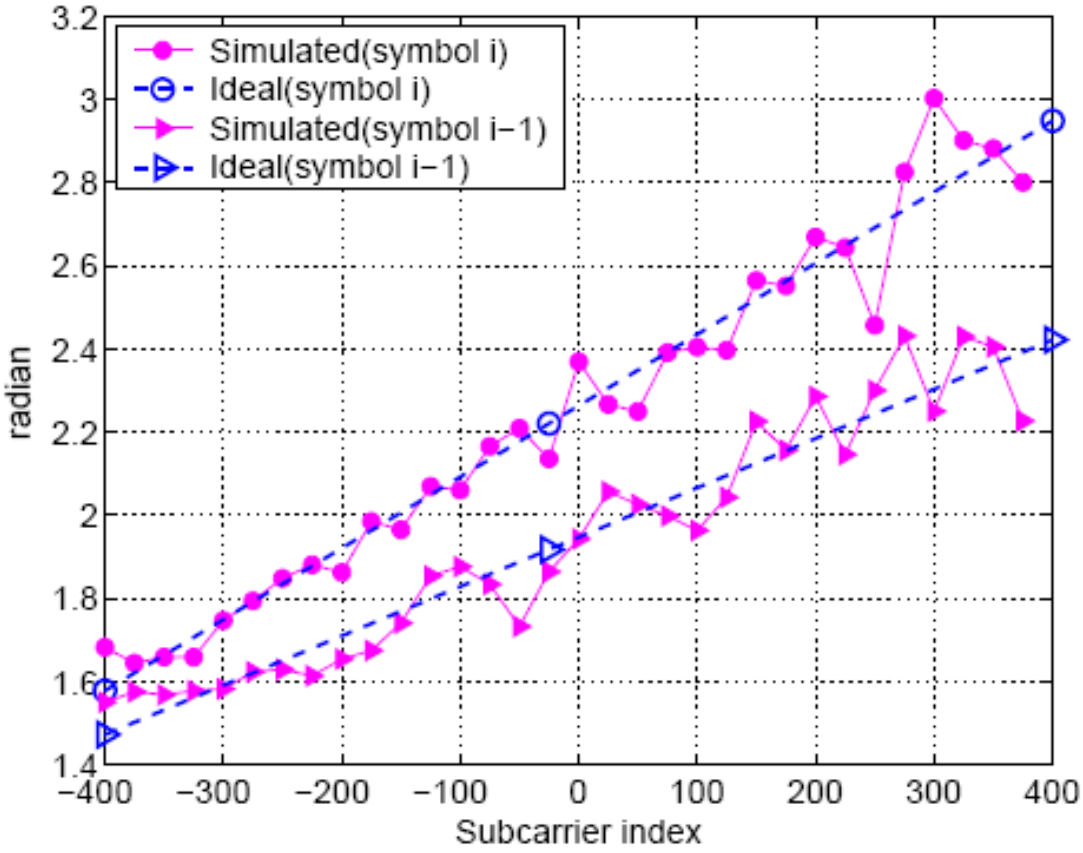
$$\Phi_{\mathrm{FDAC}}(g) = \left| \sum_{p=0}^{P-1} Z_{i,\alpha_p+g} \cdot Z_{i-1,\alpha_p+g}^{*} \right| \qquad g = 0, \pm 1, \pm 2$$

$$\hat{g} = \arg max_g \, \Phi_{FDAC}(g)$$



# Estimation for Residual Synchronization Error

- CFO
  - Identical phase shift
- SCO
  - Phase shift proportional to the subcarrier index



# Joint Linear Least Squares Estimator

- Phase difference

$$\theta_p = \measuredangle\left(Z_{i,\alpha_p} Z^*_{i-1,\alpha_p}\right), \qquad p = 0,1,\ldots,P-1$$

$\alpha_p$ : pilot subcarrier index

- CFO estimation

$$\widehat{\epsilon_f} = \frac{\sum_{p=0}^{P-1} \theta_p}{\left(2\pi \frac{N+N_g}{N}\right) P}$$

- SCO estimation

$$\hat{\delta} = \frac{\sum_{p=0}^{P-1} \theta_p \alpha_p}{\left(2\pi \frac{N+N_g}{N}\right)\left(\sum_{p=0}^{P-1} \alpha_p\right)}$$

# Degradation by CFO

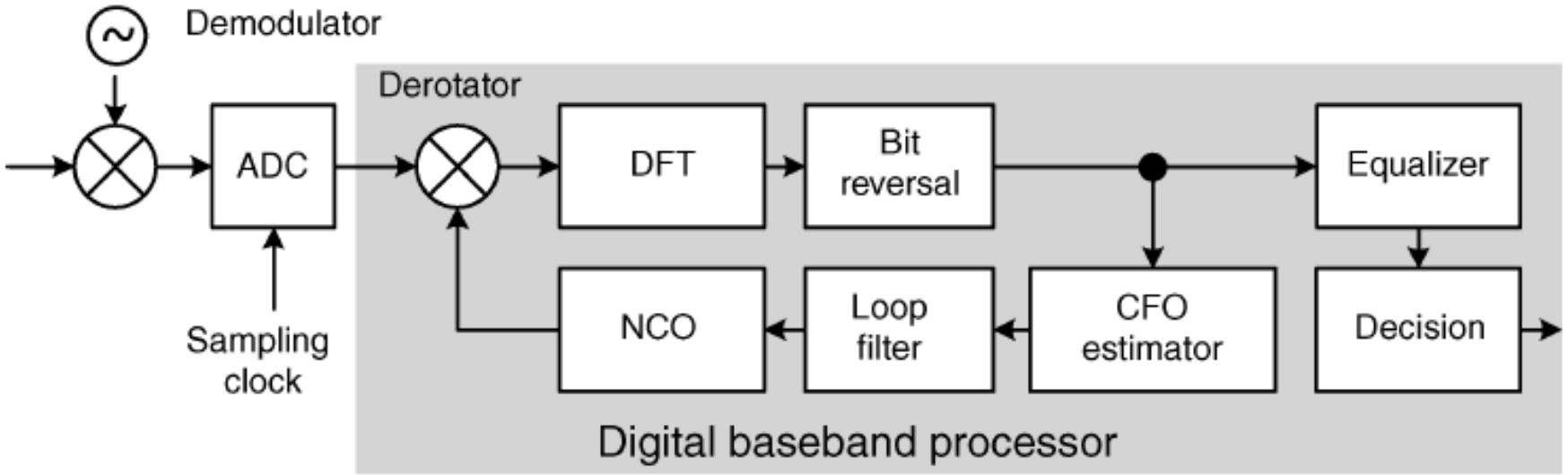
- Degradation by carrier frequency offset

$$D_{SNR} = \frac{10}{3ln10}(\pi\epsilon_f)^2\frac{E_s}{N_0}$$

  - It is better to keep residual synchronization error < 0.01 $f_{sub}$ for 64-QAM.
  - The residual synchronization error can be up to 0.03 $f_{sub}$ for QPSK.

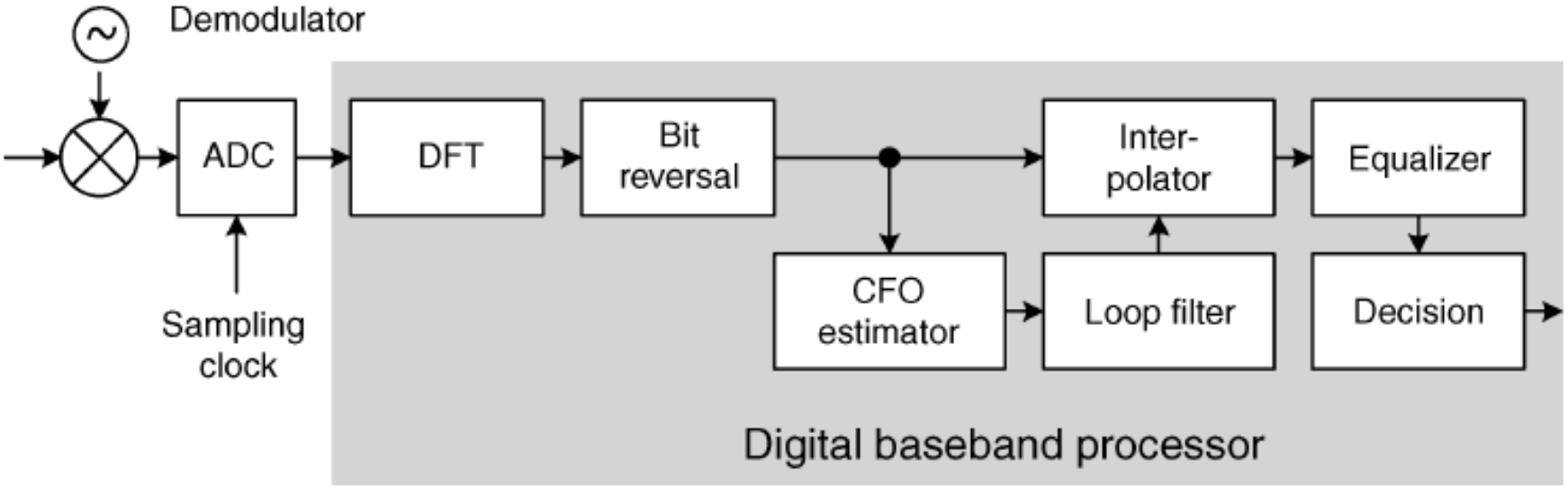
# Carrier Frequency Offset Compensation (1/2)

- Time-domain derotator



# Carrier Frequency Offset Compensation (2/2)

- Frequency-domain interpolator



# Degradation by SCO

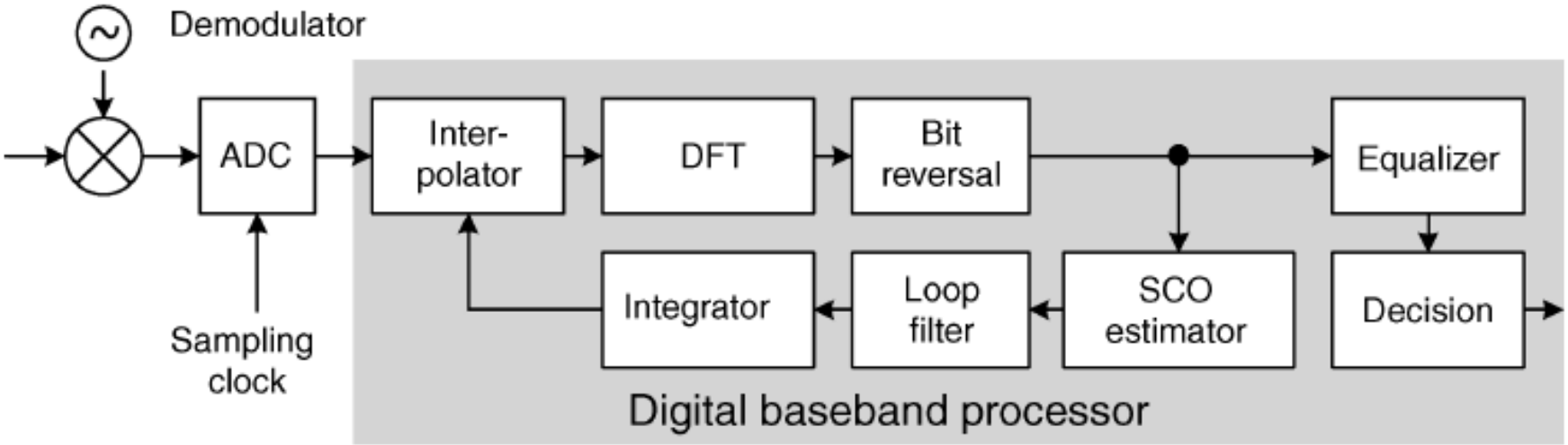
- Degradation by sampling clock offset

$$D_{SNR,k} \approx 10log\left(1+\frac{1}{3}(\pi\delta k)^2\frac{E_s}{N_0}\right)$$

  - Substantial degradation occurs at the subcarrier with larger index *k*

# Sampling Clock Offset Compensation (1/2)

- Time-domain interpolator



# Sampling Clock Offset Compensation (2/2)

- Frequency-domain phase rotator

